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MedBooking

Online Medical Appointment Booking System

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MedBooking: An Online Medical Appointment Booking System Integrating AI Chatbot and Telemedicine

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Abstract. The increasing demand for efficient healthcare services has exposed critical shortcomings in traditional appointment systems: fragmented scheduling across multiple providers, long wait times, and the absence of real-time availability information. This paper presents **Med-Booking**, a web-based medical appointment booking platform developed at FPT University as part of the SWP391 capstone project. The system provides patients with centralised scheduling, real-time slot visibility, AI-powered symptom triage, integrated telemedicine, and automated multi-channel reminders with one-click cancellation. Built on a React.js frontend, a Node.js/Express.js backend, and a MongoDB database, Med-Booking is the only platform among its Vietnamese counterparts to combine booking, telemedicine, AI chatbot, e-pharmacy, OCR prescription import, and an interactive medical map in a single application. A review of ten peer-reviewed papers from Springer (2024–2026) confirms that all major design decisions are grounded in empirically validated findings on digital health adoption and patient-centred scheduling.

Keywords: Medical Appointment Booking · Telemedicine · AI Chatbot
· Healthcare Information System · Digital Health · Patient-Centred Care

1 Introduction

Healthcare accessibility remains a global challenge. In Vietnam, patients frequently navigate fragmented booking channels — calling multiple clinics, waiting for manual confirmations, and receiving no automated follow-up — resulting in high no-show rates and poor user satisfaction. A study on MRI scheduling found that two-thirds of patients waited more than one month for an appointment, with the absence of a centralised booking point as the primary barrier [3].

Digital health platforms have emerged as a promising solution. Yet research consistently shows that perceived usefulness and ease of use are the two strongest predictors of adoption, particularly among older adults [1]. Systems that are technically capable but difficult to use fail in practice [8].

Competitive gap. Table 4 (Section 3) shows that existing Vietnamese platforms each cover at most three of six key feature areas. BookingCare offers booking and a medical map but lacks telemedicine and AI; eDoctor adds telemedicine and e-pharmacy but has no chatbot or map. No single platform combines all six capabilities.

Contribution. MedBooking addresses this gap by delivering:

1. A four-step centralised booking flow with real-time slot availability (FR-10 to FR-13).
2. An AI chatbot for symptom-based specialty recommendation (FR-33).
3. Integrated telemedicine via WebRTC video call and Socket.io chat (FR-30, FR-31).
4. Evidence-based automated reminders with one-click cancel/reschedule links (FR-16).
5. OCR prescription import and e-pharmacy integration (FR-32, FR-34).
6. A Google Maps medical facility locator (FR-35).

The remainder of this paper is organised as follows. Section 2 reviews related work. Section 3 describes the system architecture and features. Section 4 discusses alignment between the literature and our design. Section 5 concludes with future directions.

2 Related Work

2.1 Telemedicine Adoption and the TAM Model

He et al. [1] surveyed 377 older adults in China using the Technology Acceptance Model (TAM) combined with emotional design theory, analysed via structural equation modelling. *Perceived usefulness* ($\beta = 0.41, p < 0.001$) and *perceived ease of use* ($\beta = 0.35, p < 0.001$) were the strongest adoption predictors; technological anxiety had a significant negative effect. These results informed MedBooking’s UX priorities: a four-step booking flow (NFR-15), 16px minimum font size (NFR-13), and WCAG 2.1 AA colour contrast (NFR-14).

Rafi [8] conducted a mixed-methods study with 200 elderly participants in India (January–June 2025). Although 52% reported using at least one AI tool, low digital literacy (49%) and lack of trust (37%) remained major barriers. The study directly motivated MedBooking’s proxy booking feature (FR-05), which allows users to manage appointments on behalf of less tech-savvy relatives.

2.2 Centralised Scheduling and Wait-Time Reduction

Singer et al. [3] analysed MRI scheduling experiences of 557 Israeli adults; 28% found their wait of one to two months unreasonable, and two-thirds waited over a month. The root cause was the absence of a unified booking point. The study recommended a centralised hub with transparent wait-time display — the core architectural principle of MedBooking (FR-10, FR-11).

Kaluza et al. [4] modelled online booking systems using a multinomial logit and Markov decision process framework. A counterintuitive finding emerged: displaying *all* available slots can reduce bookings because users infer low demand from many empty slots. Their recommendation to expose only a controlled subset of slots directly shapes MedBooking’s slot-display strategy in FR-11.

Carlini et al. [7] evaluated an adaptive booking system for a large Italian infrastructure company, identifying quality, flexibility, and responsiveness as the three key success drivers. Their customer-centric methodology is reflected in MedBooking’s personalised chatbot flow and adaptive scheduling.

2.3 Evidence-Based Reminder Design

Selim et al. [6] ran a discrete choice experiment with 361 Australian adults. The analysis identified SMS delivery fewer than three days before the appointment ($\beta = 2.42$, $p < 0.001$), transparent cost information, and a direct cancellation/reschedule link ($\beta = 1.36$, $p < 0.001$) as the highest-valued reminder attributes, achieving a 97% acceptance rate. MedBooking’s FR-16 implements exactly these attributes: email and SMS reminders at 24 h and 1 h, each containing a one-click management link.

2.4 AI, NLP, and Clinical Decision Support

Sharma [5] reviewed AI, robotics, and NLP applications across cardiac, diabetic, gynaecological, and dermatological telemedicine. The review demonstrates that NLP enables efficient symptom triage, personalised recommendations, and virtual consultation support. This constitutes the theoretical foundation for MedBooking’s AI chatbot module (FR-33), which classifies patient-reported symptoms into the appropriate specialty with a target accuracy of $\geq 80\%$ (Goal G3).

Silva-Ferreira et al. [2] conducted a scoping review of 57 AI studies in palliative care. A key finding is that ethical and data-privacy barriers — not technical limitations — are the primary obstacles to adoption. MedBooking addresses these concerns through JWT authentication with a 24 h token expiry (NFR-05), bcryptjs password hashing with salt rounds ≥ 10 (NFR-04), and httpOnly cookie storage (NFR-07).

2.5 Stakeholder Requirements and System Quality

Ostadmohammadi et al. [9] analysed perceptions of electronic appointment systems across three stakeholder groups (10 administrative staff, 8 doctors, 18 patients) using qualitative content analysis. Problems fell into two categories: management issues (national, university, and clinic levels) and non-management issues (functional and non-functional). The study emphasises that dissatisfaction arises when non-functional requirements are under-specified. MedBooking addresses this by maintaining explicit, literature-grounded NFR sections (NFR-01 to NFR-15).

A systematic review on digital health appointment platforms [10] synthesised evidence across multiple studies and identified five critical success factors: (1) multi-channel communication, (2) real-time slot display, (3) booking in ≤ 3 steps, (4) automated confirmation and reminders, and (5) self-service cancellation. Platforms lacking these features showed a 35 % higher no-show rate. MedBooking satisfies all five criteria (FR-12 to FR-16).

3 System Architecture and Key Features

3.1 Three-Tier Architecture

MedBooking follows a standard three-tier web architecture:

- **Presentation layer:** React.js with TailwindCSS for responsive layout (375 px–1920 px), Socket.io client for real-time events.
- **Application layer:** Node.js / Express.js with JWT authentication (24 h access token, 7-day refresh token), bcryptjs, Nodemailer, and a REST API documented via Swagger.
- **Data layer:** MongoDB Atlas via Mongoose ODM, extended with Google Maps API (FR-35), Agora/WebRTC (FR-30), Tesseract.js for OCR (FR-32), and an AI/NLP service (FR-33).

3.2 Actor Model

Table 1 lists the four primary actors.

Table 1: System actors and literature basis

Actor	Role	Ref.
Patient	Books appointments; uses chatbot, telemedicine, health records	[3,6,9]
Doctor	Sets schedule; manages consultations; uses dashboard	[9]
Admin	Approves accounts; manages specialties; views statistics	[9]
Email System	External actor; executes automated notification commands	[6]

3.3 Core Functional Requirements

Table 2 presents the primary functional requirements, each grounded in the literature review.

3.4 Non-Functional Requirements

Table 3 summarises the key non-functional requirements.

Table 2: Core functional requirements with research basis

ID	Feature	Description	Ref.
FR-01	Registration	Email + password sign-up with OTP verification	—
FR-02	Login	JWT auth, remember-me session	—
FR-03	Health survey	Collects age, conditions, allergies at first login	[7]
FR-05	Relative mgmt.	Proxy booking for family members (elderly support)	[8]
FR-10	Doctor search	Filter by specialty, name, hospital, GPS location	[3]
FR-11	Slot display	Real-time slot visibility with controlled subset	[4]
FR-12	Book appointment	4-step flow: specialty → date → slot → confirm	[10]
FR-14	Cancel booking	Self-service cancellation ≥ 24 h before	[6]
FR-16	Auto reminder	Email + SMS at 24 h and 1 h; one-click cancel/reschedule	[6]
FR-30	Video call	WebRTC real-time video consultation; latency < 500 ms	[1,5]
FR-33	AI Chatbot	Symptom triage → specialty recommendation ($\geq 80\%$)	[5,2]
FR-32	OCR prescription	Tesseract.js photo import → e-pharmacy auto-fill	—
FR-35	Medical map	Nearest clinics and pharmacies via Google Maps API	[10]

3.5 Database Schema

Three primary MongoDB collections support the core workflow:

- **Users** — stores role-based profiles (`patient` | `doctor` | `nurse` | `admin`), health profiles (age, conditions, allergies), and relative lists for proxy booking.
- **Doctors** — extends **Users** with specialty, hospital, rating, biography, and a working-schedule array of time slots (each carrying an `isBooked` flag).
- **Bookings** — links patient and doctor, records date, time slot, consultation type (`in-person` | `telemedicine` | `home-care`), and status (`pending` | `confirmed` | `completed` | `cancelled`).

3.6 Competitive Analysis

Table 4 compares MedBooking against four existing Vietnamese platforms across six feature dimensions.

Table 3: Key non-functional requirements and literature basis

ID	Requirement	Ref.
NFR-01	API response time < 2 s (normal network)	—
NFR-03	Video call latency < 500 ms	—
NFR-04	bcryptjs password hashing, salt ≥ 10	[2]
NFR-05	JWT access token 24 h; refresh token 7 days	[2]
NFR-09	Responsive UI from 375 px (mobile) to 1920 px	[1]
NFR-13	Body font size ≥ 16 px (elderly accessibility)	[8]
NFR-14	Colour contrast ratio $\geq 4.5:1$ (WCAG 2.1 AA)	[1]
NFR-15	Booking flow ≤ 4 interaction steps	[10]

Table 4: Feature comparison with competing Vietnamese platforms

Platform	Booking	Telemd.	AI Bot	E- Pharm.	Map	OCR
BookingCare		—	—	—		—
eDoctor			—		—	—
Medpro			—	—	—	—
BVDHYD App		—	—	—		—
MedBooking						

4 Discussion

4.1 Alignment with the Literature

Every major design decision in MedBooking maps directly to an empirically validated finding in the literature review:

- The four-step booking flow (NFR-15) satisfies the ≤ 3 -step benchmark in [10] while maintaining a clear confirmation step.
- Proxy booking (FR-05) addresses the digital-literacy barrier for elderly users identified in [8].
- The reminder design (FR-16) implements the highest-ranked attributes from [6]: SMS, < 3 -day lead time, cost info, and one-click self-service.
- JWT + bcrypt security (NFR-04/05) responds to the ethical and privacy barriers highlighted in [2].

4.2 Identified Research Gaps

Table 5 maps gaps identified in the literature to MedBooking features.

Table 5: Research gaps addressed by MedBooking

Gap in existing research	MedBooking solution	FR/NFR
No AI chatbot integrated into the Symptom triage chatbot booking flow		FR-33
Passive reminders without self-service interaction	One-click cancel/reschedule link	FR-16
Telemedicine separated from the booking platform	Built-in video call + chat	FR-30/31
No OCR prescription integration in booking systems	Photo → e-pharmacy auto-fill	FR-32
No proxy booking for elderly or low-literacy users	Relative profile management	FR-05
Full slot display causing negative quality perception	Controlled slot subset display	FR-11

4.3 Limitations

Several limitations should be acknowledged. First, the system has not yet undergone user testing with real patients and doctors; adoption metrics remain projected from the literature. Second, the AI chatbot’s 80% specialty classification target (G3) has not been empirically validated in the Vietnamese medical context. Third, load testing for concurrent usage beyond 100 users (NFR-02) is planned but not yet executed. Finally, the OCR prescription pipeline relies on Tesseract.js, which may underperform on handwritten prescriptions.

5 Conclusion

This paper presented MedBooking, a patient-centred online medical appointment booking system developed at FPT University (Danang campus) for the SWP391 capstone course. The design is informed by a structured review of ten Springer papers published between 2024 and 2026, covering telemedicine adoption, scheduling optimisation, reminder design, AI in healthcare, and stakeholder requirements.

The key contributions are: (1) a unified platform that combines all six feature areas absent from any single Vietnamese competitor; (2) evidence-based UX and NFR decisions traceable to peer-reviewed findings; and (3) a research-gap analysis that motivates five novel features (AI chatbot, interactive reminders, integrated telemedicine, OCR, and proxy booking).

Future work will focus on:

- Usability testing with real patients and doctors in Danang.
- Benchmarking the AI chatbot’s specialty classification accuracy against ground-truth clinical data.
- Implementing the controlled slot-display algorithm proposed by [4] and measuring its effect on booking rates.

- Exploring a React Native mobile application to extend reach beyond responsive web.

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